

NOTE: Structural items will be 3D printed for cost and time effectiveness. This includes parts such as the structural body, gears, rack, etc. Items such as fasteners and screws for assembly are not included in this BOM.				
COMPUTE & CONNECTION				
ITEM	QUANTITY	COST (per unit)	WEBSITE	NOTES
ESP32 DevKit V1	1	0\$	N/A	Provided by teaching team
OV2640 Camera Module	1	1-25\$	TBD	
Breadboard	1	0\$	N/A	Easily rented
Jumper Wires + Connectors	As Needed	0\$	N/A	Myfab scrab bin
Fuses + Fuse Holder	1	1-5\$	TBD	Safety Precaution
ACTUATION & CONTROL				
ITEM	QUANTITY	COST	WEBSITE	NOTES
MG90S Servo	2	1-10\$	TBD	1 for pivoting base, 1 for driving pinion back and forth
Nerf Blaster Motors	2	13-14\$	https://outofdarts.com/collections/motors-2019	These motors would be ideal for speed, but other motors are also acceptable
Flywheels (Rubber)	2	10-25\$	https://www.studica.ca/en/2inch-compliant-wheel-12-inch-hex-40-durometer-orange	Standard wheels may be used as an alternative
TB6612FNG (Dual) Motor Driver	1	3-25\$	TBD	
Buck Converter	1	12-20\$	TBD	Step down input voltage from wall / laptop charger
Limit Switch	1	1-3\$	TBD	Adds a reference to rack and pinion system
Power Switch	1	1-2\$	TBD	
RPM Sensor	1	5-20\$	TBD	Closed loop control of flywheel speeds via PWM
INITIAL ESTIMATED COST (lower bound):		48-149\$		